

Leaning Over to Reach Far-Away Objects

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Abstract—Humanoid athleticism requires precise whole-body control under sustained load and contact, not only rapid motion. We study supported reaching: extending workspace by bracing against the environment rather than stepping. We present a sampling-based, contact-implicit MPC framework for whole-body leaning, bracing, reaching, and recovery on a Unitree H1-2 humanoid, with contact timing and forces emerging from MuJoCo rollouts rather than a prescribed schedule. The controller transfers sim-to-real, extends reach by approximately 20 cm beyond the unbraced standing workspace, sustains the loaded posture for over four minutes, and achieves 5.3 ± 0.9 cm mean steady-state error. We further characterize stability, contact loading, and end-effector precision, and analyze failure modes arising from alternative support strategies.

I. INTRODUCTION

Humanoid robots have made significant progress in perception, manipulation, and locomotion, marked by impressive demonstrations of high-speed, dynamic, and contact-rich behaviors inspired by human athletics [1, 2, 3, 4].

However, athleticism is not wholly defined by speed. Human athletic performance also depends on accurate whole-body coordination, often under sustained load: a rock climber reaching a hold or a billiards player stabilizing an extended posture. More broadly, motor-control theory has long characterized skilled movement as the coordination of many biomechanical degrees of freedom to perform physically-constrained tasks [5, 6]. A similarly complete view of athletic humanoid behavior should thus include not only explosive motion, but also motion which can establish, sustain, and release load-bearing contacts while maintaining balance and task-space precision.

One representative task is reaching distant objects. Humans can increase their reachable workspace by shifting their balance and loading onto available environmental supports rather than taking a step. While locomotion may often be the solution to an out-of-reach target, it is not always desirable or available. In industrial settings such as human-collaborative battery disassembly [7], locomotion may interrupt other agents in a workcell. In athletic tasks such as cue sports or climbing, environmental contacts may be strategically and/or geographically constrained. In this work we study supported reaching: using the upper body to brace against the environment, extend the whole-body workspace beyond the standing limit, and subsequently recover without locomotion.

We present a sampling-based, contact-implicit model predictive control (CI-MPC) framework for whole-body leaning, bracing & reaching, and recovery on a Unitree H1-2 humanoid. The resulting controller transfers sim-to-real with 73% success, extends the robot’s reachable workspace by 20 cm beyond unbraced standing, and sustains the resulting loaded posture for more than four minutes. We further study

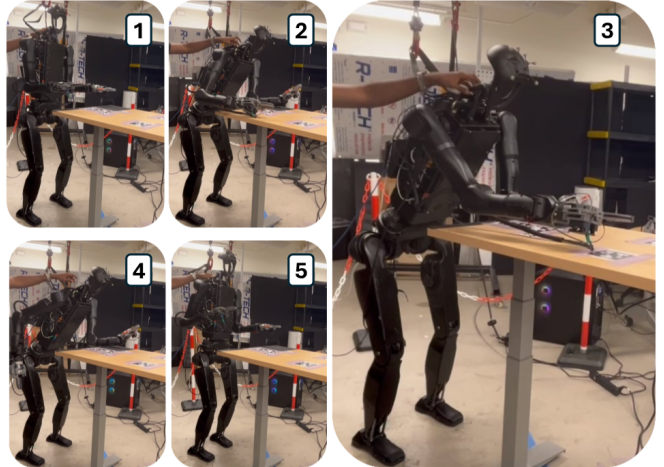


Fig. 1: Five stages of standing, leaning, leaning & reaching, recovering, and standing again on the Unitree H1-2. With sampling-based model predictive control, the robot can hold the 1 m reaching stage (frame 3), 20 cm past the max reach of the standing workspace, for over four minutes with mean target error 5.3 cm and precision 2.5 cm.

the whole-body stability, target-tracking error, and steady-state jitter of the controller.

II. RELATED WORK

a) Whole-body and multi-contact control: Humanoid whole-body control becomes particularly challenging when employing environmental contacts beyond the feet, a problem commonly studied as multi-contact planning and control (MCPC), where a controller must reason over both the robot’s whole-body motion and a set or sequence of environmental support contacts [8]. Common approaches separate these problems into two: first planning a contact sequence, and then optimizing whole-body control trajectories (and contact forces) under a priori contact constraints.

Several recent systems demonstrate the value of upper-body support contacts for humanoid manipulation. Among “classical” control methods, most relevant is SEIKO [9], which performs real-time multi-contact whole-body control on the position-controlled TALOS humanoid, including far-reaching motions that exploit explicitly planned environmental bracing contacts for support. Related work extends this controller with learned multi-support behaviors and automatic support placement for manipulation [10], while other approaches adapt contact locations or whole-body posture to improve stability during multi-contact teleoperation [11]. Among learning-based methods which instead encode whole-body behavior into reusable policies or skill spaces, most relevant to our work are: Opt2Skill [12], which learns to track dynamically feasible supported reaches generated by trajectory optimization with

explicitly-planned contacts (Crocodyl [13]), and R²S² [14], which learns a whole-body skill space for large-workspace humanoid reaching, without environmental contacts.

While these various recent works have demonstrated large-workspace and multi-contact humanoid reaching, their evaluation has emphasized task success and endpoint tracking. We provide focused characterization of supported reaching, measuring workspace extension together with whole-body stability, target-tracking precision, and steady-state jitter. Furthermore, we investigate reaching across a contact-implicit spectrum: with and without brace-encouraging objectives. Rather than planning an explicit contact sequence, retargeting a human motion, or learning to track a reference, sampling based MPC discovers and sustains environmental contact for successful reaching through planner rollouts over rigid-body simulation (MuJoCo). Autonomous discovery of contacts via CI-MPC eliminates manual, potentially limited, contact planning, and sampling-based methods are of particular recent interest due to their amenable structure for GPU-based parallelization [15].

b) MuJoCo MPC: Model predictive control is a common approach for whole-body control of legged robots. MuJoCo MPC (MJPC) [16] is a GUI-based platform interfacing physical simulation in MuJoCo, many classes of MPC planners (sampling and gradient based), and a large set of tasks and robotic embodiments. We therefore employ MJPC for ease of rapid development and iteration. The framework uses iterative linear-quadratic regulator (iLQG) methods with finite-difference derivatives, enabling real-time operation at 20-50Hz on modern hardware. Recent work has demonstrated MJPC with iterative Linear-Quadratic-Regulator (iLQR) planning on the Unitree H1 humanoid [17, 18], showing the promise of MJPC for real-time, real-world locomotion and manipulation on modern computing platforms. Our work extends MJPC deployment to sampling-based planning on the Unitree H1-2 (two more DoFs) without a motion-capture system.

III. METHODS

A. Model Predictive Control

We employ MPC within the MuJoCo physics simulator, specifically leveraging MJPC. At each control timestep, the first control u_0 is applied and the planner re-solves a trajectory optimization problem from the updated measured state over a finite-horizon T of 1s:

$$\min_{u_{0:T}} \sum_{t=0}^T c(x_t, u_t) \quad (1)$$

subject to dynamics $x_{t+1} = f(x_t, u_t, \Delta t)$, where $x_t \in \mathbb{R}^{n_x}$ is the state (positions and velocities), $u_t \in \mathbb{R}^{n_u}$ is the control (joint torques or position targets), and $c(\cdot)$ is the running cost over the finite horizon.

MJPC decomposes the cost $c(x, u) = p(l(x, u); R)$, where the base cost is defined as:

$$l(x, u) = \sum_{i=1}^M w_i \cdot \mathbf{n}_i(r_i(x, u)) \quad (2)$$

Algorithm 1 One planning iteration of sampling-based MPC

Input: control θ , state x_0 , rollouts n , horizon T

- 1: $\bar{\theta} \leftarrow \text{CENTER}(\theta)$ $\triangleright$ what to sample around
- 2: **for** $k = 1 \dots n$ **do**
- 3: $\epsilon^{(k)} \sim \mathcal{N}(0, \Sigma^{(k)})$ $\triangleright$ SPREAD
- 4: $\theta^{(k)} \leftarrow \bar{\theta} + \epsilon^{(k)}$
- 5: $u_t \leftarrow \text{spline}(\theta^{(k)}; t)$, roll out f from x_0
- 6: $J^{(k)} \leftarrow \frac{1}{T} \sum_{t < T} c(x_t, u_t)$ $\triangleright$ Eq. (1)
- 7: **end for**
- 8: $\theta \leftarrow \text{FOLD}(\{\theta^{(k)}\}, \{J^{(k)}\})$ $\triangleright$ how the winner is kept
- 9: apply $\text{spline}(\theta; 0)$; shift θ forward one step

Table I. How planners instantiate Alg. 1. σ is the sampling standard deviation; rollouts are per planning iteration.

Planner	CENTER	SPREAD	FOLD	Roll.
Predictive Sampling	θ	$\sigma = 8$	$\arg \min_k$	10
Cross-Entropy	elite mean	elite cov.	elite mean	20

This base cost is a sum of M terms, each comprising of a tunable weight $w \in \mathbb{R}_+$, a twice-differentiable norm function $\mathbf{n}(\cdot)$, and residual terms $r \in \mathbb{R}^p$. Then to modulate risk sensitivity, an exponential scalar transformation p is applied for some scalar parameter $R \in \mathbb{R}$, where $R > 0$ is risk-averse and $R < 0$ is risk-seeking.

Each residual r_i captures a specific objective (e.g., reaching a target, maintaining balance, minimizing control effort). The weights w_i are tuned to trade off competing objectives. MJPC implements this through user-defined sensor (residual) specifications in XML with corresponding weight parameters.

MuJoCo models contact and friction with a convex approximation of underlying non-convex, non-smooth contact dynamics. In lieu of analytical derivatives of the soft contact model over a given timestep $x_{t+1} = f(x_t, u_t)$, MuJoCo approximates model derivatives (dynamics Jacobians) via finite differences:

$$\frac{\partial f}{\partial u} \approx \frac{f(x, u + \epsilon) - f(x, u)}{\epsilon} \quad (3)$$

A similar expression can be constructed for $\partial f / \partial x$ and finite-difference step ϵ [19, 18]. These state and control Jacobians, and their second-order derivative Hessians, are commonly used for planning in MPC, such as iterative Linear Quadratic Regulator (iLQR) or differential dynamic programming (DDP), but often fail to find local minima (e.g., to discover contact) in CI-MPC problems due to un-differentiable contact discontinuities.

Thus, we primarily explore gradient-free sampling based planners to solve the CI-MPC task of brace contact selection and leaning. At each replanning step it draws a batch of N candidate control sequences $\{u_{0:T}^{(i)}\}_{i=1}^N$, each a perturbation of a nominal spline, rolls every candidate out through the MuJoCo model in parallel, and scores the accumulated cost $\sum_t \ell(x_t^{(i)}, u_t^{(i)})$ along each rollout. Alg. 1 describes one planning iteration of sampling-based MPC, and Tab. I compares, for didactic purposes, cross-entropy method (CEM) [20],

TABLE II: MJPC planner cost terms. $[x]_+ = \max(0, x)$; d_i^{seat} and d_i^{keep} denote the clearance of link i from the usable and physical table surfaces, respectively, and $\mathcal{B}$ is the selected brace set.

Term	Residual
Brace	$\min(\bar{d}, [d_i^{\text{seat}} - d_{\text{sat}}]_+), \quad i \in \mathcal{B}$
Keep-out	$[d_{\text{safe}} - d_i^{\text{keep}}]_+, \quad i \notin \mathcal{B}$
Trunk clear	$[d_{\text{safe}} - d_{\text{trunk}}]_+$
Reach	$p_{\text{hand}} - p^{*, \text{reach}}$
Balance	$ c_{\text{cp}} - \Pi_S(c_{\text{cp}}) $
Feet planted	$([z_L - z_0]_+, [z_R - z_0]_+, 1 - \hat{z}_L^\top e_z, 1 - \hat{z}_R^\top e_z)$
Height	$[h_{\text{min}} - (z_{\text{head}} - \frac{z_L + z_R}{2})]_+$
Regularization	$q - q_{\text{home}}; \quad \dot{q}; \quad u - q_{\text{home}}$

which we employ, and predictive sampling, an elementary yet effective method [16], w.r.t. components of Alg. 1.

Alternative approaches may use analytical derivatives of rigid-body models (Pinocchio [21]) for gradient-based trajectory optimization, such as Crocoddyl [13], which provides differential dynamic programming (DDP) solvers. While such methods offer fast convergence through exact gradients, they require more careful contact model specification and may be less robust to model mismatch than gradient-free, sampling-based approaches. Crocoddyl additionally requires explicit contact specification, limiting autonomous contact discovery and generalization (motivating Opt2Skill, which employs Crocoddyl for initial data-generation for an RL tracking policy trained for robust real-world hardware deployment [12]).

B. Task Formulation

The task consists of three sequential modes: **(1)** standing, **(2)** reaching toward an object, and **(3)** returning to standing.

Tab. II describes the cost terms used for CI-MPC. Here, $c_{\text{cp}} = c_{\text{CoM}} + 0.3 \dot{c}_{\text{CoM}}$ is the capture point, and $\mathcal{S}$ is the rectangular support region defined by the feet. Once a selected brace is within 10 mm of the table, it is treated as seated for purposes of extending the support region $\mathcal{S}$. The quantities d_i^{seat} and d_i^{keep} denote geometric clearances from the usable bracing surface and the physical table surface, respectively; the former uses a 60 mm inset from the table edge to prevent satisfying the seating objective by contacting the lip. The set $\mathcal{B}$ contains the links selected for bracing by nonzero brace weights. Thus, brace terms encourage those links to approach and seat on the table, while keep-out terms hold all unselected candidate links away from it. While this cost shape is a gray area between explicit and implicit contact planning, MJPC does not enforce a specific contact schedule and nominally requires contact discovery emergent from sampling over rolled-out dynamics. Further cost descriptions are shown in App. Tab. III.

The deployed configuration (20 rollouts over a 1 s planning horizon) plans at a median 34Hz on a laptop with an AMD Ryzen AI 9 HX 370 CPU. Preliminary testing on a desktop with an Intel i7-14700F CPU showed ≈ 24 Hz planning and increased failure. We apply thin rubber grip-tape to the wooden office table to increase the static coefficient of friction with the

anodized aluminum robot arm links from $\mu \approx 0.3$ to $\mu \approx 0.8$.

C. State Estimation

To estimate the floating-base position and velocity $x_b, v_b \in \mathbb{R}^3$, we fuse proprioceptive measurements—IMU orientation R_b and angular velocity ω_b , together with joint positions and velocities $q, \dot{q}$ —with exteroceptive position measurements from LiDAR and AprilTag fiducials. The proprioceptive estimator uses a random-walk Kalman model for base velocity,

$$v_{b,k} = v_{b,k-1} + w_k, \quad z_k = v_{b,k} + \eta_k, \quad (4)$$

where w_k and η_k denote process and measurement noise, respectively.

During standing, each planted foot provides a kinematic observation of base velocity [22]. Assuming zero world-frame velocity of contact point k ,

$$z_k = -[\omega_b \times (p_k(q) - p_b) + R_b J_k(q) \dot{q}], \quad (5)$$

where J_k is the translational contact Jacobian. Observations from the two feet are incorporated with standard Kalman updates using contact-weighted, outlier-gated measurement noise [23]. Integrating the resulting velocity estimate provides base position, but residual orientation and kinematic errors lead to long-horizon drift [22, 24]. We therefore additionally fuse LiDAR- and AprilTag-based position observations to provide an external reference and bound this accumulated position error.

IV. EXPERIMENTS

We evaluate our methods on a Unitree H1-2 in simulation (MuJoCo) and in the real-world. Detailed information regarding torque limits, simulation modeling, and hardware interface details are provided in App. A.

A. Estimation

We evaluate three state estimation methods:

1. **Internal:** IMU + leg-kinematic odometry with kinematic-chain guards,
2. **Internal + LiDAR:** the internal estimator fused with FAST-LIO [25],
3. **Internal + Fiducials:** the internal estimator fused with position observations from an external bundle of six AprilTags (36h11, 5–10 cm wide), viewed by the head-mounted Intel RealSense D435i camera.

We first isolate estimator drift by holding the robot stationary for 7 min per estimator variant, with both feet planted and the robot tethered and supported. Because the physical base remains stationary, any reported base translation reflects estimation error. Figure 3 shows the estimated horizontal base position over each trial. With internal estimation alone, the estimate exhibits sustained random-walk drift, moving approximately 0.52 m over a representative 120 s window and 1.8 m over the full trial.

Adding exteroception changes this behavior qualitatively: after an initial convergence transient, both fused estimates

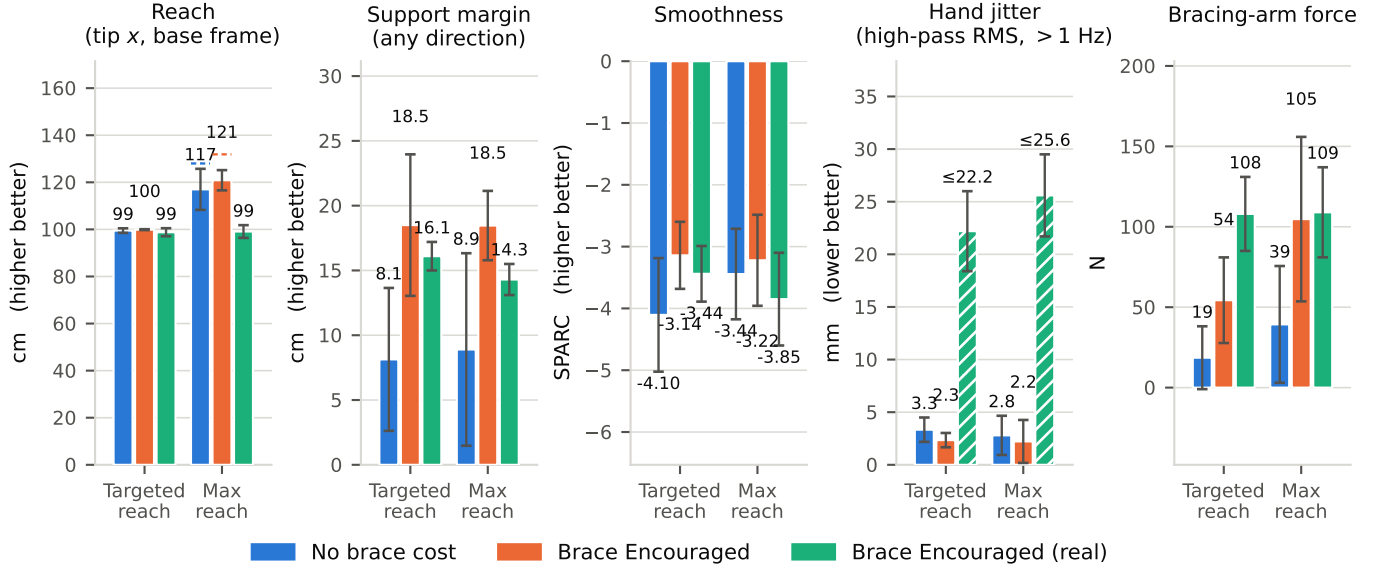


Fig. 2: Reach, stability, motion quality, and brace loading in successful runs. For each plot, results represent two reaching tasks: reaching a specific target pose (left) and reaching a maximum extent (right). Blue denotes simulation without a brace objective, orange simulation with brace-encouraging costs, and green real-world evaluation of bracing. Bars show means over successful rollouts; whiskers show half-ranges. **Support margin** is the minimum margin to the active support-region boundary; trajectory **smoothness** reports spectral arc length (SPARC, higher is smoother); **hand jitter** is high-pass RMS motion above 1 Hz; and **bracing-arm force** is the measured normal load. Real-world hand jitter bars are affected by state-estimation noise absent from simulation.

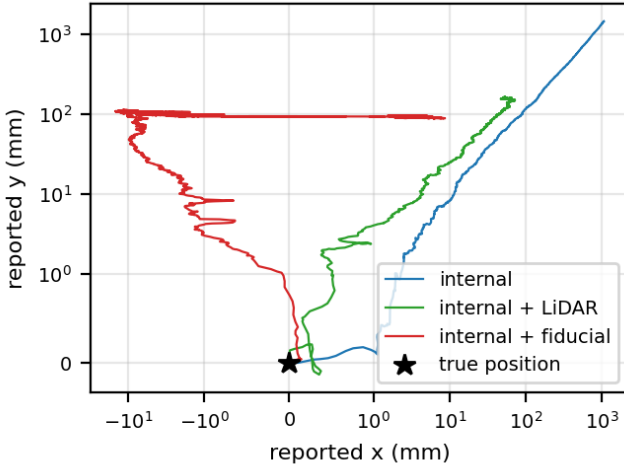


Fig. 3: XY-view of the reported base position over 7 minutes while the robot is physically stationary (star). Z-axis drift is 0.3mm over the same period. X-axis drift of the Internal + AprilTag Fiducial estimator is ± 10 mm oscillation.

remain bounded. Over 120 s windows, the LiDAR and fiducial variants remain within ≈ 12 mm and 7 mm, respectively, corresponding to a $40\text{--}75\times$ reduction in horizontal drift. Their total drifts over the full 7 min trial are 175 mm and 114 mm, respectively. Exteroceptive fusion does not improve the short-term velocity noise. Across all variants, the median reported velocity remains 8.5–8.9 mm/s. Vertical drift is negligible by comparison (< 0.3 mm over the full trial), since base height is directly constrained by the planted-leg kinematics. The

following results leverage the Internal + Fiducial estimator.

B. Leaning, Reaching, and Recovering

We evaluate two tasks: reaching to a target pose (targeted reach) and reaching to the maximum extent afforded by the planner (max reach) across two MPC formulations: with and without the bracing objective in Tab. II. With the bracing objective, in simulation, we observe $90 \pm 13\%$ success (18/20) and $(16/20)80 \pm 18\%$ success (16/20) on the targeted reach and max reach tasks, respectively, with the bracing objective. Without the objective, we report $100 \pm 11\%$ success on both tasks. On real-world hardware, we employ the brace objective because we discovered the efficacy of this brace-objective-free method (in simulation) quite late. We report $80 \pm 20\%$ success (12/15) on the targeted reach task and $73 \pm 22\%$ success (11/15) on the max reach task in real.

The remainder of this section is devoted to stability analysis of successful runs, described in whole by Fig. 2. We investigate failure in greater detail in Sec. IV-C.

a) *Emergent contact strategies:* Contact-implicit sampling does not require the realized support mode to match the one suggested by the cost function. Even with no brace objective, the optimizer frequently discovers light upper-body support on its own: in the target reach task, the planner braces the left robot gripper against the table with 19 N of force and 39 N in the max-reach task. These contacts are substantially lighter than the deliberately encouraged brace (54 and 105 N, respectively) and still yield success. Conversely, encouraging a brace does not uniquely determine a contact sequence. In rollouts with and without a bracing objective,

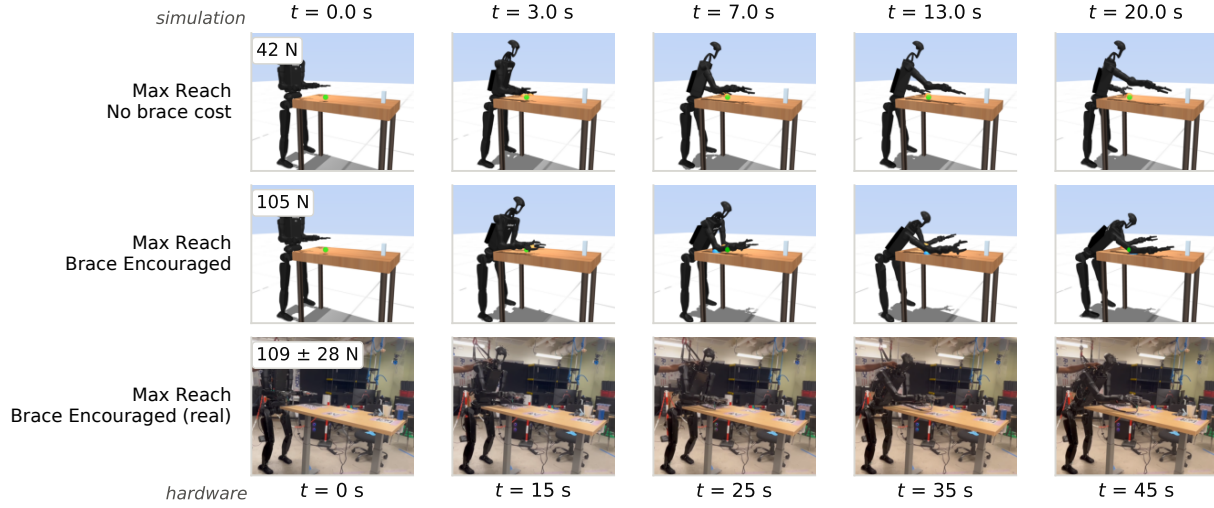


Fig. 4: Supported reaching in simulation and hardware. **Top two rows:** representative max reach rollout without (first row) and with (second row) a brace objective. Each is the median ($n = 5$) rollout by bracing-arm load. **Bottom row:** the corresponding reach, with a brace objective, on the real Unitree H1-2. Simulated rollouts span 20 s and the hardware trial 45 s due to a hardware (real-world) only stabilization phase. Labels report bracing-arm load (simulation: value for the displayed rollout; hardware: mean across trials). The hardware stands 37 cm from the near table edge, compared with 26 cm in simulation, reducing its available table-relative reach by 11 cm.

the robot sometimes places its torso on the table as it falls forward. While some successful runs leverage this additional contact, over-reliance on this contact induces large torso motor torques, triggers an emergency stop, and is the ultimate cause of all failures. Thus, the brace cost biases the optimizer toward a desired support strategy, while the final contact mode and loading remain outcomes of the simulated dynamics.

b) Reach: For the target reach task, simulation and hardware reach similar positions. The simulated reach settles at 99 ± 1 cm without a brace objective and 100 ± 0 cm with one, compared with 98.8 ± 1.7 cm on hardware. Target error, however, is larger on the robot: the simulated hand finishes 9 ± 5 and 4 ± 3 mm from the target, whereas the hardware error is 38 ± 15 mm and is predominantly Y-axis error. The discrepancy grows at the max-reach task. Simulation reaches 117 ± 9 cm without the brace objective and 121 ± 4 cm with it, while the largest demonstrated hardware reach remains 99 ± 3 cm.

c) Support margin: Bracing substantially increases the simulated support margin. At the hardware target it rises from 8.1 to 18.5 cm, and at the far target from 8.9 to 18.5 cm. The braced condition is also more consistent at the far target, where the half-range decreases from 7.4 to 2.7 cm. Hardware measurements fall within the simulated braced range at both reaches, with margins of 16.1 ± 1.1 and 14.3 ± 1.2 cm. Thus, despite the mismatch in maximum reach, simulation and hardware agree closely on the stability benefit provided by the additional support contact.

d) Bracing-arm force: In simulation, greater reach is accompanied by greater arm loading. Mean bracing-arm force increases from 19 to 39 N without the brace objective and from 54 to 105 N when bracing is encouraged. Hardware loads are 108 ± 23 and 109 ± 28 N at the two reach conditions, near the upper end of the simulated braced range. The real robot

therefore relies on a comparatively heavily loaded support contact even at the shorter target, despite attaining less reach than the simulated braced controller at the far target.

e) Smoothness and jitter: The brace has a modest effect on simulated trajectory smoothness. At the hardware target, SPARC improves from -4.10 to -3.14 ; at the far target the change is smaller, from -3.44 to -3.22 . Hardware values of -3.44 and -3.85 lie within the simulated variation, indicating comparable low-frequency motion smoothness. High-frequency hand motion shows a larger sim-to-real gap. Simulated jitter is 2.2–3.3 mm RMS and changes little with the brace objective, whereas the hardware measurements are 22.2 ± 3.8 and 25.6 ± 3.9 mm. These hardware values should be interpreted as upper bounds on physical hand motion: the measurement includes state-estimation noise, while the simulated controller receives noise-free state.

1) Target Position Accuracy: To quantify absolute end-effector precision, we command the gripper tip to a fixed Cartesian target in the table frame (0.55 m past the near edge, 0.18 m above the surface) and require a sustained 5 s hold before the pipeline proceeds to recovery. Across 10 autonomous trials, the mean placement error during the hold is 5.3 ± 0.9 cm, shown in Fig. 8. Decomposed along the table axes, the error along the reach (depth) axis is unbiased (0.0 ± 1.7 cm), with a repeatable lateral offset of 3.9 ± 1.5 cm and a height offset of 1.9 ± 1.1 cm; the per-trial precision is ≈ 2.5 cm RMS.

C. Failure Analysis

Successful and failed trials separate primarily in how they load and maintain the brace. We analyze the 15 max-reach hardware trials (11 successes, 4 failures). During bracing, both groups move the CoM beyond the foot-only support polygon (Fig. 5). Successful trials gradually recover to a

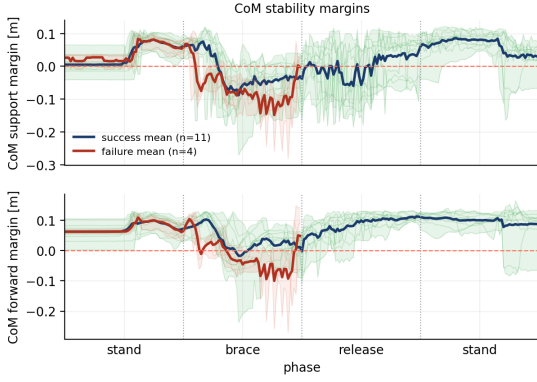


Fig. 5: CoM support margin over hardware trials. Time is normalized to the commanded phase boundaries for each trial. Top: minimum distance from the horizontal CoM projection to the **foot-only support polygon**. Bottom: front-back distance to its front edge. Negative values lie outside the foot support region.

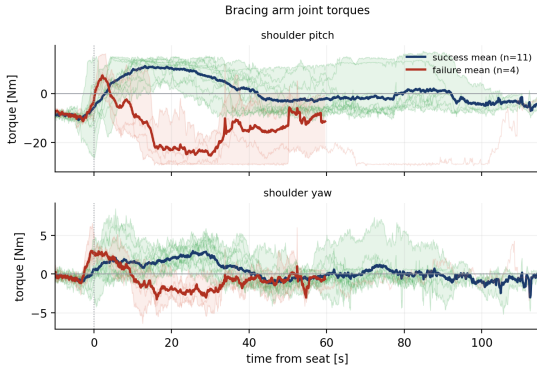


Fig. 7: Bracing-arm shoulder torques. Trials are aligned to the measured seating instant ($t = 0$, dotted line). Curves show shoulder pitch and yaw torques for successful and failed trials.

positive margin up through brace release. In failed trials, the robot slowly falls forward after the initial brace, reaching approximately -0.14m overall and below -0.10m relative to the front edge before triggering an emergency stop.

The corresponding contact-force profiles further distinguish successful and failing loading strategies (Fig. 6). Successful trials load the forearm progressively, reaching approximately 125N on average, with per-trial peaks of $151 \pm 24\text{N}$, and unload gradually during recovery. Failed trials instead develop approximately 150N near brace onset and subsequently maintain a more variable $100\text{--}150\text{N}$ load until collapse.

Shoulder torque and position measurements most precisely diagnose failure (Fig. 7). Successful trials sustain approximately $+10\text{Nm}$ of shoulder-pitch torque while braced, whereas failures reach -20 to -25Nm , reflecting configuration ($+0.14 \pm 0.08$ versus $-0.45 \pm 0.10\text{rad}$). Shoulder-yaw torque remains within approximately $\pm 5\text{Nm}$ in both cases. To provide some physical intuition, this difference reflects the initial brace placing the shoulder link behind (success) or ahead (failure) of the elbow, and currently is best attributed to sampling variance in the current planner configuration.



Fig. 6: Reconstructed brace-contact force. Normal contact force is estimated quasi-statically from measured bracing-arm joint torques through the contact Jacobian transpose. The reconstruction gives 4.3N RMSE against ground-truth force in simulation. Time is normalized to the commanded phase boundaries.

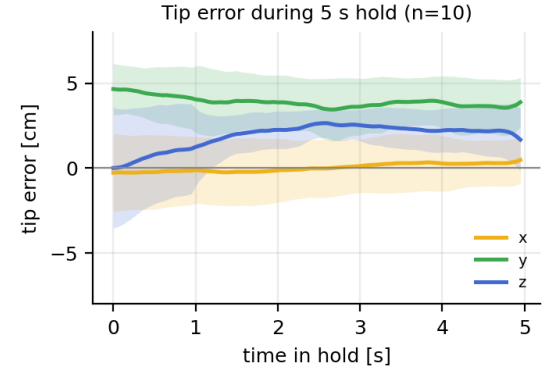


Fig. 8: Gripper tip tracking error during the sustained hold. The depth (x) component is unbiased ($0.0 \pm 1.7\text{cm}$) while the lateral (y) and height (z) components carry offsets of $3.9 \pm 1.5\text{cm}$ and $+1.9 \pm 1.1\text{cm}$.

V. DISCUSSION

Our results show that sampling based CI-MPC can discover, load, sustain, and release an environmental support contact on a H1-2 without an explicit contact schedule. The brace objective acts as a bias rather than a constraint. Simulation and hardware agree closely on support margin and low-frequency smoothness, while hardware brace loads lie near the upper end of the simulated range, suggesting that the MuJoCo rollout dynamics of MuJoCo MPC capture the essential mechanics of supported reaching.

The failure analysis yields a compact mechanical signature: failures are not falls at brace onset but a gradual forward migration after contact, associated with an extended-elbow support geometry, opposite shoulder-pitch torque, and abrupt rather than progressive loading. Every element of this signature is proprioceptive and appears seconds before collapse, well within the horizon at which the robot could still intervene.

Future work include closing the precision bias with better online fiducial calibration, extending the framework to object retrieval for varying payload mass while under brace, generalizing across support surfaces, inclinations and heights, and

migrating sampling to GPU rollouts to raise replanning rate.

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APPENDIX

A. Robot, Workcell, Actuation

Robot	Unitree H1-2, 27 actuated joints (12 leg, 1 torso, 14 arm), ≈ 75 kg floating base: $n_q=34$, $n_v=33$, $n_u=27$
Hands	MAGPIE [26] parallel grippers, with unactuated joints
Plant	MuJoCo, 2 ms timestep, Newton solver, <code>implicitfast</code> integrator
Servos	position, $k_p=200$ legs / 150 hip yaw / 40 arms
Table	1.18×0.595 m, face $z=0.985$ m, near edge $x=0.450$ m <code>solref="0.08 1", solimp="0.85 0.95 0.001 0.5 2", priority=1</code> (80 ms contact time constant)
Target	By default: (0.905, -0.235 , 1.098) m with left arm brace and right hand reach
Sites	MuJoCo contact sites placed at elbow, forearm, wrist pad, palm; feet 6D at the soles

a) *Torque*: We specify a conservative joint torque clamp enforced by a low-level command filtering node in ROS 2 and DDS (original torque limits specified in [27]):

Joint	$\bar{\tau}$	Joint	$\bar{\tau}$	Joint	$\bar{\tau}$
Hip Yaw	54	Knee	270	Shoulder Pitch/Roll	28.8
Hip Pitch	117	Ankle Pitch	48.6	Shoulder Yaw, Elbow	13.0
Hip Roll	180	Ankle Roll	32.4	Wrist Roll/Pitch/Yaw	8.6
Torso Yaw	36				

Units N m. MuJoCo `mjcf` model files specify `forcerange` constraints for the sampler.

B. Sampling Cost Terms

The sampling controller uses a single objective throughout the horizon. The intended brace mode is softly encoded through the weights on the elbow, forearm, and palm seating (contact-force generating) terms. A nonzero brace weight encourages the corresponding link to approach and seat on the table. links not selected for bracing are instead kept away from the surface by the Table Keepout term. Contact itself is determined by the MuJoCo dynamics during rollout, so the optimizer is free to realize the intended interaction at different times—or not realize it at all. Unless otherwise stated, the selected brace mode is {elbow, forearm}.

TABLE III: Sampling-controller cost terms. SA denotes the smooth-absolute activation $\sqrt{r^2 + p^2} - p$, and Q a quadratic activation. Geometric gaps are measured relative to the table surface; brace seating residuals use a 60 mm inset from the edge of the table.

Term	Residual	Norm (p)	w
Brace Elbow	Upper-arm gap, saturating at contact	SA (0.05)	300
Brace Forearm	Forearm gap, saturating at contact	SA (0.05)	300
Brace Palm	Gripper gap, saturating at contact	SA (0.05)	0
Table Keepout	$\max(0, 50 \text{ mm} - \text{gap})$, links off-mode (3)	SA (0.02)	200
Trunk Clear	$\max(0, 50 \text{ mm} - \text{gap})$, torso and pelvis	SA (0.02)	1500
Reach	$p_{\text{hand}} - p_{\text{target}}$ (3)	SA (0.05)	400
Balance	Capture point outside support box (2)	SA (0.02)	300
Feet Planted	Foot rise and sole tilt, per foot (4)	SA (0.02)	500
Height	$\max(0, 1.20 \text{ m} - h_{\text{head-fee}})$	SA (0.05)	100
Posture	$q - q_{\text{home}}$ (27)	Q	1.0
Joint Vel.	$\dot{q}$ (27)	Q	0.01
Control	$u - q_{\text{home}}$ (27)	Q	0.1

a) *Support region*: Before bracing, the balance term is computed using the support polygon of the feet. Once a planned brace is observed within 10 mm of the table, the support region is extended toward that contact. This allows the controller to exploit the additional support created by the brace: retaining balance computation derived from a feet-only support region after contact would penalize the center-of-mass excursion from leaning.

b) *Seating, not pressing*: The brace residuals are positional cost terms which encourage contact establishment rather than a prescribed contact force. They penalize separation from the surface while approaching the table, and go to zero once the corresponding link is “seated”. The objective rewards reaching the surface but does not reward pushing farther into it.